# Financial Machine Learning Part 0: Bars

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The opening instalment of a series applying López de Prado's Advances in Financial Machine Learning to real datasets, starting with bars. A hands-on, code-first tour of the four standard bar types — time, tick, volume and dollar — built on Bitcoin (XBT) trade data, computing a VWAP per bar and comparing the series.

The memorable payoff: plotting tick bars against time bars on the same Bitcoin data surfaces a roughly 10% flash rally and flash crash that were hidden inside the time-bar representation — events that are either a mean-reversion opportunity or a slippage cost depending on the strategy.

Each bar type is motivated from the shortcomings of the previous: time bars oversample quiet periods; tick bars fix the activity mismatch but treat a 1,000-contract trade the same as ten 100-contract trades; volume bars fix that but are distorted by price level; dollar bars sample on traded value and are most robust to splits and large price moves. A friendly, adaptable code template and a good primer before imbalance/run bars or mlfinlab.

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